BHAVIN SANGANI

Graduate Data Analyst

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Available for full-time opportunities in the UK | MSc Data Analytics graduate (Merit), De Montfort University | Graduate Route visa application planned, subject to eligibility

PROFESSIONAL SUMMARY

MSc Data Analytics (De Montfort University, course completed Sep 2026) with a 6-month Data Analyst internship using SQL, Power BI and Python for KPI reporting and dashboards. Built a tested Python and PostgreSQL ETL pipeline and Power BI dashboard on NHS England A&E data (107 million attendances, reconciled to published totals with zero difference), forecast attendances within 1.72% MAPE, and traced late deliveries across 180,000+ order lines to unrealistic promised lead times. Looking for a graduate or junior Data Analyst role where accurate, well-validated reporting drives decisions.

TECHNICAL SKILLS

Analysis & Reporting: SQL (CTEs, window functions), Excel (Pivot Tables, VBA), KPI Reporting, Root Cause Analysis, EDA

Visualisation & BI: Power BI (DAX, Power Query), Tableau, Matplotlib, Seaborn

Programming: Python (Pandas, NumPy, Scikit-learn, Statsmodels, XGBoost, SQLAlchemy), R

Data Engineering & Databases: ETL Pipelines, Data Modelling (Star Schema), Data Quality Checks, PostgreSQL, MySQL, Microsoft SQL Server, SQLite

Statistics & Modelling: Regression, Classification, Time-Series Forecasting (Holt-Winters), Class Imbalance (SMOTE), Data Validation & Reconciliation

Tools: Git / GitHub, Docker, pytest, Jupyter Notebook, SAS Studio

PROJECTS

NHS England A&E 4-Hour Performance Analysis

Python • SQL (SQLite) • Pandas • Statsmodels • Matplotlib | github.com/BSsangani/nhs-ae-performance

- Combined 48 monthly NHS England A&E files (Apr 2022 – Mar 2026; 221 providers, 107 million attendances) and reconciled them to published totals with zero difference, after finding 48 hidden 'TOTAL' rows that would have double-counted activity.
- Showed England's 4-hour performance was 74.9% in 2025/26; March 2026 reached 77.1%, just short of the 78% target, while 12-hour waits from decision to admit rose 39% to 570,931.
- Forecast monthly A&E attendances with Holt-Winters: 1.72% MAPE on 12 unseen months, beating a seasonal-naive baseline (2.44%).
- Benchmarked University Hospitals of Leicester: improved from 54.7% to 62.8% (+8.1 points), but still below the England average.

NHS A&E Data Pipeline & Power BI Dashboard

Python • PostgreSQL • SQL • Power BI (DAX, Power Query) • pytest • Docker | github.com/BSsangani/nhs-ae-data-pipeline

- Built an incremental, idempotent ETL pipeline loading 48 monthly CSVs into PostgreSQL (raw → staging → mart), using a file-hash log so re-runs load only new files.
- Automated 7 data quality checks (duplicates, negative values, reconciliation to published totals); critical failures stop the run, and 4 pytest tests include an end-to-end idempotency test.
- Modelled a star schema (9,623-row fact table with date, provider and region dimensions) feeding a 3-page Power BI dashboard: national overview, regional comparison and trust explorer.
- Wrote DAX measures (CALCULATE, DIVIDE, RANKX) for 4-hour performance, gap to the 78% target, year-on-year change and provider rankings.

Supply Chain Late Delivery Analysis & Prediction

Python • SQL (SQLite) • Pandas • XGBoost • Matplotlib • DataCo dataset | github.com/BSsangani/supply-chain-late-delivery

- Analysed 180,519 order lines (54,854 orders) with SQL KPI queries: 57.3% of orders were late; First Class was late 100% of the time (promised 1 day, always 2) and Second Class 79.8%.
- Found the root cause was unrealistic promised lead times, not region or market (all five markets 56.8–57.8% late), and recommended resetting service levels by shipping mode.

- Built an XGBoost model using only information known at order time, with a time-based split: 0.78 ROC-AUC (vs 0.746 for Logistic Regression); orders in the highest-risk 20% were late 97.9% of the time vs 32.6% for the lowest 20%.

Customer Churn Prediction

Python • Pandas • Scikit-learn • XGBoost • SMOTE • Matplotlib | github.com/BSsangani/customer-churn-prediction

- Predicted churn for 7,043 telecom customers; tuned Random Forest and XGBoost with GridSearchCV, with XGBoost best at 78% accuracy and 0.83 ROC-AUC on a held-out test set.
- Handled a 26.5% churn rate by applying SMOTE to the training set only, and identified contract type as the strongest driver (42.7% churn on month-to-month vs 2.8% on two-year contracts).

WORK EXPERIENCE

Warehouse Operative (Part-time) – Oakland International, Coalville **Aug 2026 – Present**

- Pack frozen food orders in a fast-paced cold-storage warehouse at Oakland International's Coalville site.
- Use handheld scanners to scan and track items accurately during packing.
- Follow food safety and health & safety procedures when handling frozen products.

Packaging Operative (Part-time) – Sangam Paneer Manufacturing, Leicester **Sep 2025 – Jul 2026**

- Operated packaging machinery in a fast-paced food manufacturing environment.
- Followed quality and health & safety standards to keep products safe and correctly packed.
- Worked within a diverse team while studying full-time for an MSc.

Career Break – Preparation for Postgraduate Study in the UK **Jul 2024 – Aug 2025**

- Prepared for postgraduate study abroad, including English language exams, university admissions and visa applications.

Data Analyst Intern – World Standards Scientific LLP, Surat, India **Jan 2024 – Jun 2024**

- Wrote SQL (MySQL) queries to extract, join and validate KPI data drawn from multiple business data sources.
- Built and maintained Power BI dashboards giving the business a regular view of its KPIs and growth trends.
- Cleaned and prepared data in Python (Pandas), checking for errors and inconsistencies before it reached reports.
- Maintained existing reports and queries so business reporting stayed accurate and up to date.

EDUCATION

MSc Data Analytics – De Montfort University, Leicester **Sep 2025 – Sep 2026**

- Course completed September 2026; final results awaited.
- Key modules: Data Analytics Infrastructure, Big Data Applications, Data Mining Techniques and Methods, Project Proposal, Planning and Project Management, PGT Project.
- Dissertation: How Data Quality Affects Energy Demand Forecasting in UK Smart Grids, using NESO electricity demand data.

MSc Information Technology – Parul University, India **Sep 2022 – Jun 2024**

- CGPA 7.52 / 10 (First Class Equivalent).
- Core subjects: Software Development, Database Management Systems, Web Technologies and Computer Networks.
- Completed a master's project covering system design, implementation and testing for a real-world problem.

Bachelor of Computer Applications (BCA) – Uka Tarsadia University, India **Sep 2019 – Jun 2022**

- Core subjects: programming (C, C++, Java, C#), database management, web development and computer networks.
- Completed a final-year project designing, building and testing a software application.

CERTIFICATIONS

- RDBMS PostgreSQL Training – IIT Bombay (Certificate ID 34034231L4)
- PHP and MySQL Training – IIT Bombay (Certificate ID 3403423QCJ)
- Foundations: Data, Data, Everywhere – Google, via Coursera
- Python – HCL / GUVI (Partner of Google)